

Final Team Project

Start Assignment

- Due Jun 22 by 11:59pm
- Points 180
- Submitting a text entry box, a website url, a media recording, or a file upload
- Attempts 0
- Allowed Attempts 3

Introduction

The final project for this course will assess the development of an end-to-end statistical analysis. In Module 2, you were assigned to a group by your instructor. You and your teammate(s) have worked closely to find a dataset, cleaned and prepared it for analysis, performed EDA, and produced a model. You will then produce a final technical report discussing the results of your analysis and the validity of your model. It is required that you and your team use GitHub as a code hosting platform to manage version control and collaboration during this project. It is also necessary to create and add a [README \(https://sandiego.instructure.com/courses/31183/files/4431503?wrap=1\)](https://sandiego.instructure.com/courses/31183/files/4431503?wrap=1) file to your GitHub repository. It is *recommended* that you follow the [PEP 8 !\[\]\(c3d993ca47bfe2a953c700506ce31fa0_img.jpg\)](https://pep8.org/) (<https://pep8.org/>) – Style Guide for your Python code in the final project.

Project Timeline

- Module 2 (by the end of Week 2): The course instructor will group students into teams of two to three members.
- Module 4 (by the end of Week 4): Each team will select and introduce a dataset. The team representative will submit the "Team Project Status Update Form."
- Module 7 (by the end of Week 7): Each team will submit the following deliverables for the course project in the final week:
 1. **Technical Report:** One **PDF** document with your final technical report. This should describe your preparation and analysis of the data, discuss the final model selection, and describe the statistics behind your final model selection.
 2. **Team Presentation:** One 8-10 minute video presentation by **all** team members. **Do not exceed 10 minutes.** Submit one **MP4** file. This should be presented as a business presentation of your analysis to a non-technical audience. You will present your analysis and findings in a way that is understandable to any non-technical executive or business leader. This presentation should include **one slide** to showcase your collaborative efforts; you will create a presentation slide **highlighting each team member's individual name** and **their contributions** to the final project work and deliverables.

****It is critical to note that no extensions will be given** for any of the final projects' due dates for any reason, and final projects submitted after the final due date will not be graded.

Project Datasets

- You are responsible for finding a dataset for the final team project in this course. Several free data resources are provided by the [UCI Machine Learning Repository](https://archive.ics.uci.edu/datasets)  (<https://archive.ics.uci.edu/datasets>).

Requirements

- Divide the work equally between the team members for the following steps, and everyone needs to code and review the code.

For your technical report document:

- You should include the following sections in your report:
 - Introduction
 - Data Cleaning/Preparation
 - Exploratory Data Analysis
 - Model Selection
 - Model Analysis
 - Conclusion and Recommendations.
- Include an appendix with the output of your code from a technical notebook (i.e., Jupyter Notebook).

For your video presentation:

- Give an 8-10 minute presentation of your analysis aimed at a non-technical audience of business leaders.
- Each team member should participate in the presentation equally.

Project Deliverables and Submission Format

- Prepare and submit your Final Technical Report in **PDF** format on Canvas. You will use the naming convention **Final-Project-Report-Team-Number.pdf** (e.g., Final-Project-Report-Team-1.pdf)
- Prepare a recorded video presentation of your project using a screencasting tool, such as Zoom, to record your screen and provide a voice narration.
 - Ensure that the sound quality of your video is good and that each member presents an equal portion of the presentation.

- Export the video file to an **MP4** format.
- You may use any recording software you wish. View the [Recording Video Presentation and Submission Guidelines \(https://sandiego.instructure.com/courses/31183/files/4432261?wrap=1\)](https://sandiego.instructure.com/courses/31183/files/4432261?wrap=1) for AAI Students.
- You will use the naming convention **Final-Project-Presentation-Team-Number.mp4** (e.g., Final-Project-Presentation-Team-1.mp4).
- Only **ONE** member of your team will need to submit these deliverables. Please note that you must submit each deliverable separately, meaning you will upload a PDF of your written report with appendix, and an MP4 file (and/or a link) for your video submission.
- **NOTE:** Team members may not get the same grade on the Final Team Project, depending on each team member's level of contribution.
- You will submit the peer evaluation form individually below in Assignment 7.1.

This assignment has **Turnitin** [↗\(https://help.turnitin.com/integrity/student/canvas/assignments/submitting-an-assignment.htm\)](https://help.turnitin.com/integrity/student/canvas/assignments/submitting-an-assignment.htm) enabled for submissions, which means that your instructor will obtain a Similarity Report that identifies specific parts of your writing that may indicate a high level of matching to external content. You are strongly encouraged to review your work without penalty by activating the **Draft Coach extension in your Google Docs** [↗\(https://help.turnitin.com/integrity/student/draft-coach/using-draft-coach.htm\)](https://help.turnitin.com/integrity/student/draft-coach/using-draft-coach.htm) prior to submitting your work for final grading.

***AI-assisted tools such as ChatGPT, Gemini, and GitHub Copilot should be used to enhance your learning, not replace it. If you use any AI tools in your assignments, you *must* explicitly disclose, cite, and **explain** their contributions (e.g., comments in code, footnotes in reports). Submitting AI-generated code and answers without sufficient understanding, modification, and justification is not acceptable. Proper attribution is essential to maintain academic integrity across all courses and will help position you for future success in this field. Review the full [Use of Generative AI Tools policy \(https://sandiego.instructure.com/courses/31183/assignments/syllabus\)](https://sandiego.instructure.com/courses/31183/assignments/syllabus) in the syllabus.

Groupwork Requirements

The final **teamwork** project is an essential component of our courses in the AAI program. This project is representative of the kind of collaborative project you may work on during your *career and in real-world projects*. A significant portion of your final grade is drawn from your participation in this final group project, so you are strongly encouraged to work within your team and ensure that each team member contributes *equally* to the final project deliverables. Team members should plan to have clear and ongoing communication with their team and engage with the project and its deliverables each week. Lack of participation and engagement with both your team and your final project can result in a failing grade for the course.

While these are the expectations for the project work, if you do experience difficulty with project advancement or challenges with team dynamics, contact your instructor for assistance promptly. If you are unable to perform the project as a team, contact your instructor to explore pursuing the entire final project independently.

To understand how your work will be assessed, view the scoring rubric below.

Click the **Start Assignment** button above to submit your assignment.

Final Team Project Scoring Rubric (AAI-500)

Criteria	Ratings					Pts
Technical Report 50%	90 pts Exceeds Expectations Exceptional. Technically correct, clearly written, and well-organized report. All analyses are appropriate, correctly executed, and thoroughly interpreted. Assumptions and limitations are clearly discussed. Demonstrates strong graduate-level rigor and depth. This level of performance should be uncommon.	75 pts Meets Expectations Strong. Analysis is largely correct and clearly presented. Minor issues in clarity, depth, or justification may be present. Interpretation is mostly sound but lacks polish or depth in places.	60 pts Approaches Expectations Adequate: Report demonstrates partial correctness. Key analyses or explanations are missing, weakly justified, or unclear. Results may be interpreted mechanically without sufficient reasoning.	40 pts Below Expectations Marginal: Major analytical, interpretive, or structural problems. Methods are incorrect or poorly chosen. Report is difficult to follow or incomplete.	0 pts Non-Performance Non-Performance: Report not submitted or does not meaningfully address the task.	90 pts
Python Code 30%	54 pts Exceeds Expectations Exceptional: Code runs without errors, is well-structured, readable, and reproducible. Clear variable naming and comments explain key steps. Code reflects sound software and	42 pts Meets Expectations Strong: Code runs correctly and produces expected results. Some sections lack clarity, comments, or optimal structure, but logic is mostly sound.	30 pts Approaches Expectations Adequate: Code runs with issues or requires manual fixes. Limited comments or unclear structure. Reproducibility or readability is weak.	15 pts Below Expectations Marginal: Code contains major errors, does not fully run, or is very difficult to interpret. Demonstrates limited understanding of implementation.	0 pts Non-Performance Non-Performance: Code not submitted or does not run at all.	54 pts

Criteria	Ratings					Pts
	data science practices at the graduate level.					
Team Presentation 20%	36 pts Exceeds Expectations Exceptional: Clear, professional, and well-paced presentation (8–10 minutes). Content is accessible to a non-technical audience and clearly summarizes the project goals, methods, and findings. Demonstrates strong preparation.	28 pts Meets Expectations Strong: Presentation is mostly clear and professional. Minor timing, clarity, or delivery issues. Content is accurate but may be unevenly explained.	20 pts Approaches Expectations Adequate: Presentation covers core material but lacks clarity, polish, or focus on key takeaways. May be overly technical or poorly summarized.	10 pts Below Expectations Marginal: Presentation is poorly structured, outside time limits, or difficult to follow. Key results or explanations are missing or incorrect.	0 pts Non-Performance Non-Performance: Presentation not delivered or largely incomplete.	36 pts
Total Points: 180						